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*Original Clinical Science*

# Automated Bone Cancer Detection Using Deep Learning on X-Ray Images

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**Abstract**

In recent days, bone cancer is a life-threatening health issue that can lead to death. However, physicians use CT-scan, X-rays, or MRI images to recognize bone cancer, but still require techniques to increase precision and reduce human labor. These methods face challenges such as high costs, time consumption, and the risk of misdiagnosis due to the complexity of bone tumor appearances. Therefore, it is essential to establish an automated system to detect healthy bones from cancerous ones. In this regard, Artificial intelligence, particularly deep learning, shows increased attention in the medical image analysis process. This research presents a new Golden Search Optimization along with Deep Learning Enabled Computer Aided Diagnosis for Bone Cancer Classification (GSODL-CADBCC) on X-ray images. The aim of the GSODL-CADBCC approach is to accurately distinguish the input X-ray images into healthy and cancerous. This research presents the GSODL-CADBCC technique that leverages the bilateral filtering technique to remove the noise. This method uses the SqueezeNet model to generate feature vectors, and the GSO algorithm efficiently selects the hyperparameters. Finally, the extracted features can be classified by improved cuckoo search with a long short-term memory model. The experimental results demonstrate that the GSODL- CADBCC approach attains highest performance with an average accuracy of 95.52% on the training set data and 94.79% on the testing set data. This automated approach not only reduces the need for manual interpretation but also minimizes the risk of diagnostic errors and provides a viable option for precise medical imaging-based bone cancer screening.

**Keywords**

computer aided diagnosis, golden search optimization, medical imaging, deep learning, bone cancer

## Introduction

Muscles in the body are attached to bones to support movement. Spongy bone marrow is found within the fibrous tissue that makes up bone ligaments. A bone tumor starts off as normal cells and develops into malignant cells.[1,2] It might destroy bone tissues and bone will be weaker. For cancer patients, early identification is the only thing that increases their chances of survival. Differential diagnoses of bone cancer are based on the assessment of the age of the patient and traditional radiographs.[3] The plain radiograph is the most useful analysis for distinguishing such cases. Although positron emission tomography (PET), computed tomography (CT), magnetic resonance imaging (MRI), and bone scans are more sensitive in detecting bone cancer, these modern imaging modalities are time-consuming and expensive. In addition to demographic data like the type of matrix, patient's age, location, presence of periosteal reaction, radiographic appearance of the tumours involving size, cortical destruction, and margin are other important clues assisting the radiologist to distinguish indolent from aggressive bone cancers.[4] Since bone tumors are relatively uncommon and exhibit various appearances, only a few radiologists have developed sufficient expertise to make definitive diagnoses. Consequently, the accuracy of bone lesion analysis by radiologists is generally low, which can lead to frequent misdiagnoses that adversely affect patient outcomes.[5]

Early classification and detection are done to assure effective medication of bone cancers.[6] Radical surgical resection for primary malignancy is only feasible if the

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cancer is detected in its early stages. Radiotherapy, protected weight-bearing or surgical augmentation is required for preventing fractures around osteocytes cancers.[7] Though plain radiographs were commonly utilized for routine screening for bone cancers, higher rate of misdiagnoses upon visual inspection was stated, as bone cancers show common ambiguous features and various morphologies. Developments in artificial intelligence (AI) approaches are presenting novelties in the analysis of medical data.[8] Higher-level NNs like DL, which resemble the human brain, can solve complex problems that lower-level AI cannot. Convolutional neural network (CNN) techniques produce better results while evaluating complex paradigm health care images. The ability of deep learning to understand 2D medical images is comparable to that of a typical human expert in the field. Numerous studies have demonstrated exceptional results in the classification or diagnosis of diseases using endoscopy, CT, MRI, microscope, ultrasound, and plain radiography.[9,10] In this case, bone cancer is identified using AI technology. Human error, time, and expense are significantly reduced in medical practices when the AI-related classification technique performs well.

The study presents a new Golden Search Optimization with Deep Learning Enabled Computer Aided Diagnosis for Bone Cancer Classification (GSODL-CADBCC) on X- ray images. The presented GSODL-CADBCC technique makes use of bilateral filtering (BF) technique to remove the noise. This approach, called GSODL-CADBCC, produces feature vectors by utilizing the SqueezeNet model and the hyperparameters are proficiently chosen by the GSO algorithm. Finally, the extracted features can be classified by improved cuckoo search (ICS) with a long short-term memory (LSTM) module. An extensive set of experiments were performed to study the performance of the GSODL-CADBCC method on medical imaging datasets.

## Related Works

Shabbir et al[11] provide a way for using MRI to predict the malignancy of bone cancer. The MRI scans in the datasets have diagnoses that have been verified through histological analysis. Two pre-trained residual CNNs will classify the images taken from the T1 and T2 MRI datasets. With the aid of pre-trained CNN, the algorithm converges faster during training, and it trains on smaller data. Liu et al[12] showed that using patient clinical characteristics and standard tumor radiographs, machine learning and deep learning fusion processes are validated and developed for the classification of intermediate, malignant, and benign bone tumors. The ML approach may improve the capacity to classify bone cancers by combining information from radiographs and clinical features. The studies lack focus on optimizing computational efficiency and utilizing more readily available imaging modalities such as X-rays. It shows dependency on MRI scans which are more expensive and less accessible than X-rays.

Wang et al[13] used a DL technique for automatically counting cells in colour bone marrow microscopic images. The presented algorithm makes use of a Feature Pyramid Network and a Fast RCNN to create a model that considers different illumination levels and is accountable for the stability of the colour component. The empirical result shows that the presented model is comparable to certain sophisticated systems. Pathologists can easily operate the system with the help of a user interface. The study focused primarily on cell counting, not on the classification of bone cancer.

Calin et al[14] presented a quick object-based categorization approach that makes it easier to comprehend bone scintigraphy images. Firstly, the full lambda-schedule technique along with an edge-based segmentation technique has been applied for identifying the objects in the bone scintigraphy, and the spatial and textural features of the object were evaluated. Next, a group of objects was chosen as training datasets derived on visual inspection of an image. Then were allocated to different stages of radionuclide accumulation prior to implementing the data classifications with the help of SVM and KNN classifiers. This study does not integrate advanced DL techniques for classification. Georgeanu et al[15] present a CNN architecture in addition to image processing techniques for detecting and classifying bone MRI scans into benign or malignant tumors. With the help of the transfer learning technique, the performance of both pre-trained CNN frameworks namely ResNet-50 and VGG-16 models were compared. It shows dependency on MRI scans which are more expensive and less accessible than X-rays. Moreover, exploring and comparing more pre-trained models could provide a broader understanding of which architectures are most effective for this task.

Han et al[16] assessed the effectiveness of DL classifiers for prostate cancer patients bone scans. Two distinct 2D CNN frameworks have been introduced: (1) tandem architecture incorporating the entire body as well as local patches, given as "global–local unified emphasis" (GLUE), and (2) whole body–based (WB). The study does not integrate other relevant clinical data into the DL models such as serum PSA levels or other imaging findings. Incorporating these additional data sources could potentially improve diagnostic performance. Eweje et al[17] designed a DL approach that could discriminate benign and malignant bone cancers with patient demographics and MRI. Image-related models were produced with the help of EfficientNet-B0 framework and an LR mechanism was trained through lesion position, patient age, and sex. At last, a voting ensemble model was constructed.

Although the study includes external validation, the dataset from external sources was limited. Broader validation across diverse populations and institutions is necessary for greater generalizability. Table 1 shows a thorough comparison of literature surveys.

## The Proposed GSODL-CADBCC Model

In this research, an effective GSODL-CADBCC approach was introduced for automated bone cancer classification on X-ray images. The presented GSODL-CADBCC technique encompasses BF with noise removal, GSO along with SqueezeNet consisting of feature extraction, LSTM classification, and ICS related hyperparameter tuning. Figure 1 Exhibits the working principle of GSODL-CADBCC technique.

### *Image Preprocessing Using BF Technique*

Firstly, the GSODL-CADBCC technique annihilate the noise in the X-ray images with the help of BF technique. The BF algorithm is employed in the GSODL-CADBCC model to preprocess X-ray images by reducing noise while preserving important structural details like edges. This step is crucial because medical images, such as X-rays, often contain noise that can obscure critical features needed for accurate diagnosis. BF effectively removes noise from images, which is essential for enhancing the clarity and quality of medical images. Noise can significantly affect the performance of feature extraction and classification algorithms. Unlike traditional smoothing techniques that blur edges, BF maintains sharp edges by considering both the spatial closeness and the intensity difference of pixels. This property is particularly important in medical imaging, where edges delineate anatomical structures and regions of interest. BF algorithm is a nonlinear filtering technique that completely considers spatial and pixel value information and lastly forms a compromise processing on the image.[18] In medical images, especially in X-rays, the regions of interest (such as bones and tumors) tend to be narrow and elongated compared to random noise, which is usually more dispersed and lacks structure. This characteristic allows BF to differentiate between true edges and noise. The results of the BF algorithm are evaluated in the equation (1).

$$i_{new}(x,y) = \frac{\sum_{a=x-c}^{x+c}\sum_{b=y-c}^{y+c} i(a,b)\omega(a,b)}{\sum_{a=x-c}^{x+c}\sum_{b=y-c}^{y+c} \omega(a,b)} \tag{1}$$

Where $i_n(x-y)$ denotes the pixel value obtained afterwards applying BF algorithm. $C$ indicates the window size that affects the computation of $(a,b)$, $(a,b)$ shows the pixel value of specific point in the window, and $\omega(a,b)$ embodies the value computed by two Gaussian functions. The initial Gaussian function is evaluated using equation (2),

$$\omega_s(a,b) = e\left\{\frac{(a-x)^2 + (b-y)^2}{2a\sigma^2}\right\} \tag{2}$$

Where $(a,b)$ signifies the initial Gaussian function and value is controlled by variable $\sigma s$. The second Gaussian function is evaluated using equation (3).

$$w_v(a,b) = exp\left\{\frac{((a,b)-(x,y))^2}{2\sigma^2}\right\} \tag{3}$$

Where $(a,b)$ represents the second Gaussian function.

The results of the BF algorithm are evaluated with the above equations (1)–(3) by computing a new pixel value using a weighted average of surrounding pixels. The weight $\omega(a,b)$ combines spatial closeness and intensity similarity which ensures that similar and nearby pixels have more influence. The spatial weight $\omega_s(a,b)$ depends on the distance between pixels, while the intensity weight $\omega_y(a,b)$ relies on intensity differences. This approach effectively reduces noise while preserving edges that are crucial for medical images where structural details are important. The final pixel value is obtained by normalizing the weighted sum of pixel values to enhance image clarity and usability.

**Table 1.** Comparison of Literature Survey.

| Authors | Imaging | Dataset Size | Models | Algorithms |
|---|---|---|---|---|
| Georgeanu et al.[11] | MRI | 39 MRI (malignant and benign tumors) | DL | ResNet50 |
| Liu et al.[12] | Bone tumors | 643 patients and 982 radiographs | DL | XGBoost, inception V3 |
| Wang et al.[13] | White blood cells in bone marrow images | 609 white blood cell images | DL | Faster RCNN, feature pyramid network (deep CNN) |
| Georgeanu et al.[15] | MRI | 166 Thigh bones | DL | VGG-16, ResNet-50 |
| Han et al.[16] | Bone scan | Positive: 2991, Negative: 6142 | DL | 2D-CNN |
| Eweje et al.[17] | MRI | Benign lesions: 582, malignant lesions: 478. | ML and DL | Ensemble (EfficientNet and LR) |

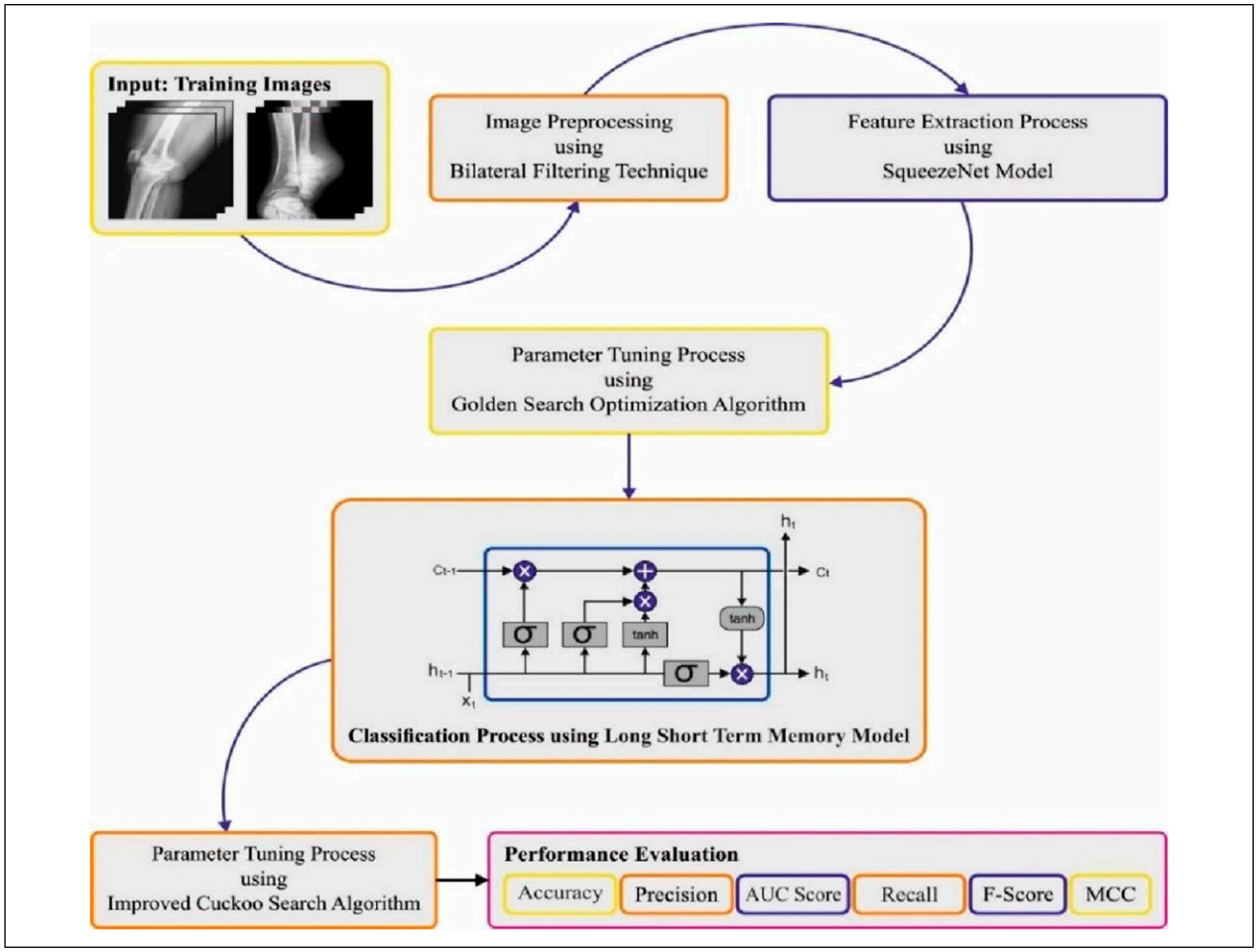


**Figure 1.** Workflow of GSODL-CADBCC model.

## *Feature Extraction Using Optimal SqueezeNet Model*

During this stage, the SqueezeNet model received the preprocessed image as input in order to produce feature vectors. CNN typically consists of pooling, convolutional, and fully connected (FC) layers.[19] Firstly, convolutional and pooling layers are used to extract the features. Then, the feature map in the convolution layer is transformed into 1*D* vector. Next, the output layer classifies the input images. The network lowers the square divergence between the classifier and predictive outcomes and changes the weighted parameter through BP. The neurons in each layer are arranged in 3D height, width, and depth where depth determines the channel count of input images or amount of input feature map, and width and height identifies the size of the neuron. The convolution layer encompasses multiple convolution filters, which extract features in the image with a convolutional model. To create the final feature map and exclude local features, the current layer's convolution filter convoluted the input feature map. Next, the nonlinear feature map was realized by the activation function. The subsampling or pooling layers are final convolution layers. It employs a downsampling method with a particular value as outcomes in a specific area. Figure 2 Shows the architecture of SqueezeNet model as follows:

The parameter count for VGGNet and AlexNet optimize, the SqueezeNet architecture is developed with minimal parameters but maintained accuracy. In SqueezeNet, the fire element develops a significant component and is detached to Squeeze and Expand the architecture. The 1 × 1 convolution layer obtained substantial consideration. Consequently, accomplishing linear combination of many feature maps and data integration on the channels. When the output and input channel count is greater, then the convolution kernel develops greater. Next, add 1 × 1 convolution to every inception system that reduces the input channel count, and the convolution kernel variable and complex function were decreased. Finally, add a 1 × 1 convolution layer to improve the number of channels as well as the feature extraction. A

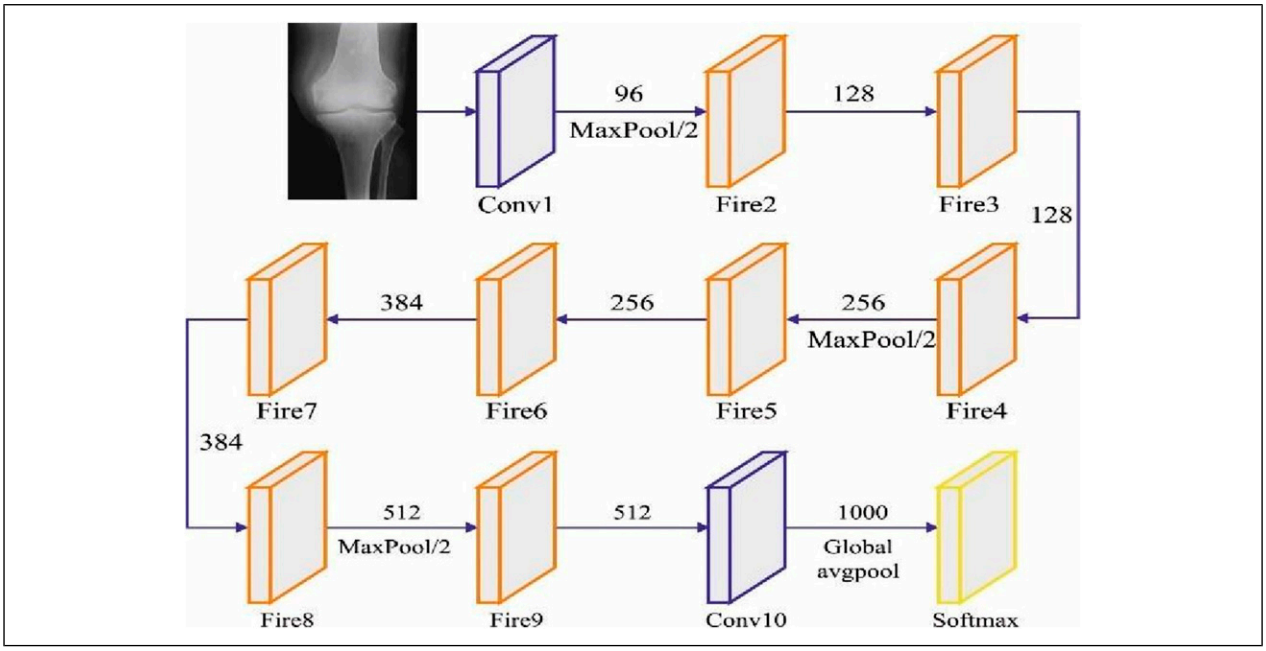


**Figure 2.** Structure of SqueezeNet.

higher activation graph was presented to the convolution layer once the sampling reduction technique is delayed, hence the highest activation graph reserves further data and provides improved efficacy of the classifier.

To improve the efficiency of the SqueezeNet module, the GSO is used. GSO includes exploitation and exploration stages and provides a balance between two conflicting capabilities. The model comprises three major phases namely initialization, evaluation, and updating the existing population. The proposed GSO steps are given in the following[20]:

$$O_i = lb_i + rand \times (ub_i - lb_i); i = 1, 2, \ldots, N \quad (4)$$

In equation (4), $O_i$ present the position of $i$th objects in the search range. Furthermore, $ub_i$ and $lb_i$ denotes the lower and upper boundaries of the object, correspondingly. During assessment, the initial population is assessed by using objective function and the objects with good fitness values are considered as $Ogbest_i$. In the golden change phase, the object is sorted based on the fitness and the object with worst fitness is altered by the random solution. In every iteration, the object was moved towards the better solution with the help of step size operators ($St_i$). $St$ equation comprises three phases. The initial part characterizes the preceding value of step size that is multiplied by the transform (T) operator that is repeatedly reduced for balancing global and local search space. The next part presents the distance among the present location of *i-th* objects and their personal better location attained by the cosine of random number within [0,1]. The last portion signifies the distance between *i-th* object's present location and the better location attained, multiplied by the sine of randomly generated numbers amongst [0,1]. In the initial iteration, $St_i$ would be randomly generated and upgraded through subsequent formulas as follows[20]:

$$\begin{aligned} S(t+1) = T.S(t) + C_1.\cos(r_1).(O_{best_i} - x_i(t)) \\ + C_2.\sin(r_2).\left(O_{gbest_i} - x_i(t)\right) \end{aligned} \quad (5)$$

In Eq. (5), $C1$ and $C2$ indicate a random integer within [0,1], $r1$ and $r2$ denote some random number within [0,1], $O_{best_i}$ denotes the best prior location attained by the i- th objects and $T$ indicates transfer operator that is used to transforms from exploration search to exploitation search for supervising the balance between global and local search in earlier and later iterations and improving the performance. $T$ is a reducing function and is estimated as

$$T = 100 \times \exp\left(-20 \times \frac{t}{t_{max}}\right) \quad (6)$$

where $t_{\max}$ indicates the maximal amount of iterations. In all the iterations, the approach progresses by regulating the distance that every object proceeds in all the dimensions of the problem space. Equation (6) demonstrates that the step size is a stochastic parameter and might enable the object to follow a broad cycle in the problem space. A reasonable interval is presented for controlling this oscillation and to prevent divergence and

explosion, to clamp the object movement based on the following expression:

$$-St_{imax} \leq St_i \leq St_{max} \tag{7}$$

In equation (7), $St_{i\max}$ indicates a selected maximal movement permitted, which determines the maximal change one object might undergo in its positional coordinate during the iteration as follows:

$$St_{imax} = 0.1 \times (ub_i - lb_i) \tag{8}$$

Finally, the object proceeds to the global optimal in the search range as follows:

$$(t+1) = O(t) + St_i\,(t+1) \tag{9}$$

## Bone Cancer Classification

For automated recognition of bone cancer, the LSTM model is used. LSTM is an innovative structure in RNN that is frequently employed to handle time series tasks. Traditional RNNs suffer from the vanishing gradient problem, where gradients become too small for effective learning during backpropagation. LSTMs mitigate this issue through their gating mechanisms which enable the network to learn and propagate gradients over long distances. In comparison to conventional FFNN, LSTM has memory ability and allows the data stream to continue to flow inside the network.[21] It is capable of linking the prior data to the existing problem. However, the past status data is useful for deciding the present state of equipment. In medical imaging, especially for tasks like bone cancer detection, it is crucial to capture the sequential nature of the image slices. Each slice of an X-ray can be considered a step in a sequence, and LSTMs are adept at learning from these sequences. LSTM has four major components, viz, output gate, input gate, forget gate, and cell state. The data of the cell state is updated by using three gates that automatically make LSTM add or remove data to the neural network. LSTMs are capable of learning complex patterns and dependencies in the data, which improves classification accuracy. With respect to bone cancer detection, the model can better distinguish between healthy and cancerous bone structures by leveraging temporal information and dependencies within the image slices. LSTMs can retain information over long sequences, which is beneficial for medical image analysis where the context from previous slices can provide essential clues for accurate diagnosis. LSTMs maintain a cell state that can preserve important information across many time steps. The use of input, output, and forget gates allows the network to decide which information to keep and which to discard enhances the model's ability to focus on relevant features. Forget gate decides how much data is forgotten from the prior cell state $C_{t-1}$ and it is mathematically modeled as:

$$f_t = \left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \tag{10}$$

$h_{t-1}$ and $x_t$ characterize hidden state at $t-1$ time and input feature at t time, correspondingly. In the study, the input feature is the output of CNN. Every component of the feature map is regarded as the input of LSTM in one moment.

$W_f$ and $b_f$ denote the weight parameter and bias term of forgetting the gate layer.

$\sigma$ shows the sigmoid function. The output of $f_t$ value lies between 0 to 1.

1 shows the data of the cell state is fully retained,

whereas 0 indicates the data has dropped out completely.

Input gate determines how much of newly learned data is added to the present cell state $C_t$:

$$C_t = tanh(Wc \cdot [h_{t-1}, x_t] + bc) \tag{11}$$

It is evaluated by nonlinearly mapping prior hidden states and existing input features.

Following, the input gate selects partial data from $C_t$ as follows:

$$i_t = (W_i \,.\, [h_{t-1}, x_t] + b_i) \tag{12}$$

The meaning of the parameter in equation (12) is the same as in equation (10). Combining equations (10)–(12), the present cell state $C_t$ is evaluated by:

$$C_t = f_t \odot C_{t-1} + i_t \odot C_t \tag{13}$$

The present cell state is made up of two terms, viz, $t \odot C_{t-1}$ and $i_t \odot C_t$. The initial one signifies the filtered data afterwards being rejected, and the last one represents the recently generated feature data. The final gate, viz, output gate, defines which part of the cell state is going to output. The last output of hidden layer $h_t$ is evaluated by:

$$o_t = W_o \,.\, [h_{t-1}, x_t] + b_o \tag{14}$$

$$h_t = O_t \odot \tanh(C_t) \tag{15}$$

From the above equations, it is clearly understood that the above mentioned three nonlinear gates regulate the data flow in and out of the LSTM network. Furthermore, the hidden state $h_t$ has every historical state data from time 0 to $t$. The time series data is useful for constructing precise health indicators.

Finally, the ICS algorithm fine-tunes the hyperparameter of the LSTM module to achieve maximum accuracy. CS is derived from the lifestyle of a bird family named cuckoo. The special lifestyle of this bird and its characteristics in breeding and egg laying are the primary

motivation for the improvement of new evolutionary optimization techniques.[22] CS begins with an early population that has matured cuckoo and their eggs. Every matured cuckoo has its egg laying radius (ELR) and lays eggs in its ELR in host bird habitat. Next, some eggs that are less like host bird eggs will be discarded. The original CS algorithm can address various optimization challenges due to its exploration and exploitation capabilities. However, to improve its performance for specific problems, some modifications are necessary. The original CS algorithm efficiently balances exploration (searching new areas) and exploitation (refining existing solutions). However, for complex or specific optimization problems, this balance might need adjustment. The performance of the CS algorithm depends on parameters like step size and probability of discovery.

Survived cuckoo population will be migrated to the best location. The first cuckoo based on the location in environments is separated into some clusters and lastly, every cuckoo will be migrated towards the better cuckoo in the better cluster with $\lambda$ percentage and with the deviation of $\alpha$ radians. Once each position of the cuckoo became closer, the process ends. Being closer means that this environment has maximum food source and fewer eggs will demise. But the original CS is capable of resolving a number of challenges, by shifting some of its operations, and based on the problems, we could enhance its overall performance. In the proposed model, local search technique is added termed "Simulated Annealing" and few operations of CS are to enhance it. The outcomes demonstrated that the modification employed on the CS enhanced the over zperformance. To enhance the overall performance of the CS algorithm, we propose several modifications: adaptively decreasing the Egg egg-laying radius over iterations, adjusting the number of eggs based on fitness, and implementing a repair mechanism for out-of-bound solutions. Additionally, integrating local search with Simulated Annealing to refine the best solutions and using clustering and migration to leverage collective intelligence. These enhancements improve exploration and exploitation balance, ensure feasible solutions, and accelerate convergence, making the ICS more effective for complex optimization problems in applications like the GSODL-CADBCC model. The changes are given below:

*Adaptive ELR.* At the highest iteration, the estimated solution should be altered less than the prior iteration, and the size of ELR should be decreased. It shows that eggs aren't laid farther away from the parents. In the presented method, the size of ELR is decreased while the amount of iterations is improved.

*Adaptive Number of Eggs.* The number of eggs of every cuckoo is randomly defined, between the minimum and the maximum number of eggs. In the presented method, it is potential that a weaker cuckoo might lay more eggs when compared to the stronger one. This might decrease the convergence speed. In the presented technique, the number of eggs of every cuckoo is set based on the cost value:

$$eggNum_i = minEggNum + \left[ (fitness_i - fitness_{min}) \times \frac{maxEggNum - minEggNum}{fitness_{max} - fitness_{min}} \right] \tag{16}$$

where, $fitness_{\min}$ = The minimal fitness in the present population, $fitness_{\max}$ = The maximum fitness in the existing population, $fitness_i$ = the fitness value of the existing cuckoo. Result shows that this development increased the convergence speed for this type of problem.

*Repair the Malformed Cuckoo.* In the original CS, few components of the habitation vector of cuckoo might surpass the search range of the problem. In such cases, these elements are returned to the search range of the problem. For instance, when the search range is $[-M, M]$, a component with value of $M + \alpha$ would be $M - \alpha$ after the repair step.

*Apply a Local Search for a Better Solution.* In all the iterations, to enhance the better solution, we applied the AE algorithm on the better cuckoo of all the clusters. Once the better cuckoo at every cluster gets enhanced, remaining cuckoos are affected by these improvements because of the Migration step. The ICS technique not only acquires a fitness function to attain high efficiency of the classification and also defines a positive integer to symbolize the good performance of the solution candidate. The decline of classification error rate was referred to as FF.

$$fitness(x_i) = \frac{number\ of\ misclassified\ samples}{Total\ number\ of\ samples} \times 100 \tag{17}$$

**Algorithm 1:** Pseudocode of the ICS

**Begin:**
i. Initialization
ii. Evaluate the cost of every cuckoo.
iii. Allocate few eggs between $\max_{num}$ and $\min_{num}$ to all the cuckoos.
iv. Evaluate the ELR for all the cuckoos and define the location of every egg of cuckoo based on the parent cuckoo.
v. If some components of the habituated vector of cuckoo exceeded the predetermined range then repair these components.
vi. If number of present cuckoos in the population is better than $\max_{num}$, then demise few weak cuckoos.
vii. Define cuckoo society by $k$-means clustering and find the better cuckoo in all the clusters.
viii. Employ a local search algorithm on the better cuckoo.
ix. Move every cuckoo of every cluster towards the better cuckoo in that cluster.
x. Define egg laying radius for all the cuckoos.
xi. When the ending condition is not fulfilled then go to (iii).
xii. The better cuckoo is the better solution.
**End**

## Results and Discussion

The bone cancer classifier outcomes of the GSODL-CADBCC method are tested with a bone cancer X-ray dataset. The dataset holds 200 images with two classes, as provided in Table 2. Figure 3 Shows a few sample images of the dataset.

Figure 4 Shows the confusion matrix generated by the GSODL-CADBCC model on the bone cancer classification process with distinct sizes of training set (TRS) and testing set (TSS). The outcomes demonstrated that the GSODL-CADBCC technique has accurately categorized the cancerous and healthy bone images.

**Table 2.** Bone Cancer Dataset.

| Class | No. Of Images |
|---|---|
| Cancerous bone | 100 |
| Healthy bone | 100 |
| Total no. Of images | 200 |

The bone cancer classification outcomes of the GSODL- CADBCC method on 80:20 of TRS and TSS are exhibited in Table 3 and Figure 5. It is noticed that the GSODL-CADBCC technique has effectively identified the cancerous and healthy bone images effectively. For instance, on 80% of TRS, the GSODL-CADBCC model has reached average *accubal* of 95.52%, *precn* of 95.76%, *recal* of 95.52%, *Fscore* of 95.60%, *AUCscore* of 95.52%, and MCC of 91.28%. At the same time, on 20% of TSS, the GSODL-CADBCC model has resulted in average *accubal* of 94.79%, *precn* of 94.79%, *recal* of 94.79%, *Fscore* of 94.79%, *AUCscore* of 94.79%, and MCC of 89.58%.

The bone cancer classification results of the GSODL-CADBCC model on 70:30 of TRS and TSS are demonstrated in Table 4 and Figure 6. It is noticed that the GSODL- CADBCC model has identified the cancerous

**Figure 3.** Sample X-ray images.

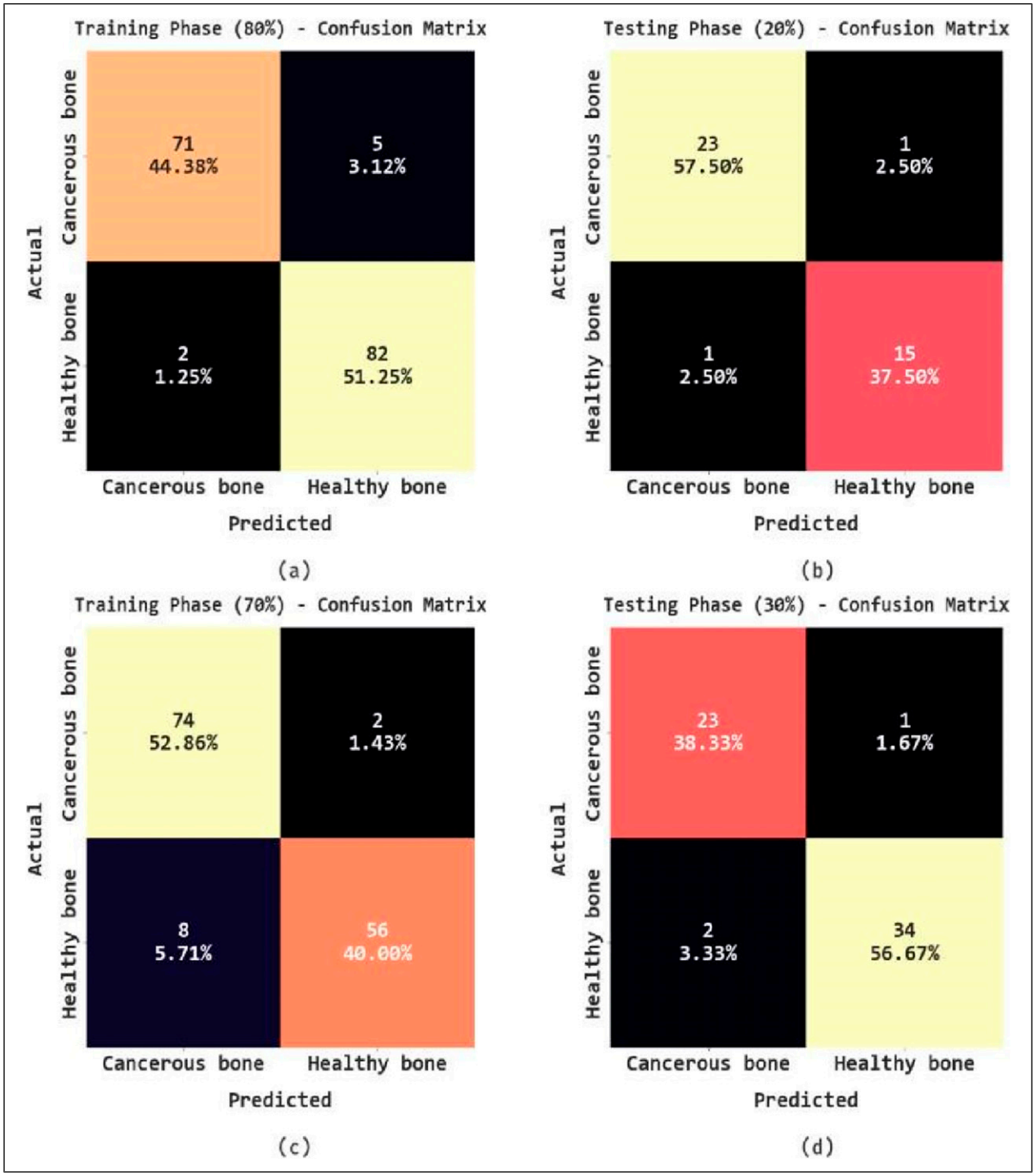


**Figure 4.** Confusion matrices of GSODL-CADBCC model (A) 80% of TRS, (B) 20% of TSS, (C)70% of TRS, (D) 30% of TRSS.

and healthy bone images effectually. For example, on 70% of TRS, the GSODL-CADBCC method has reached average *accubal* of 92.43%, *precn* of 93.40%, *recal* of 92.43%, *Fscore* of 92.74%, *AUCscore* of 92.43%, and MCC of 85.83%. However, on 30% of TSS, the GSODL-CADBCC model has resulted in average *accubal* of 95.14%, *precn* of 94.57%, *recal* of 95.14%, *Fscore* of 94.83%, *AUCscore* of 95.14%, and MCC of 89.71%.

The training accuracy (TAC) and validation accuracy (VAC) study of the GSODL- CADBCC technique is depicted in Figure 7. It can be easily noticeable that the GSODL-CADBCC model accomplishes improved values

**Table 3.** Bone Cancer Classification Results of GSODL-CADBCC Model on 80:20 of TRS/TSS.

| Class Labels | ***Accubal*** | ***Precn*** | ***Recal*** | ***Fscore*** | ***AUCscore*** | MCC |
|---|---|---|---|---|---|---|
| Training phase (80%) | | | | | | |
| Cancerous bone | 93.42 | 97.26 | 93.42 | 95.30 | 95.52 | 91.28 |
| Healthy bone | 97.62 | 94.25 | 97.62 | 95.91 | 95.52 | 91.28 |
| Average | 95.52 | 95.76 | 95.52 | 95.60 | 95.52 | 91.28 |
| Testing phase (20%) | | | | | | |
| Cancerous bone | 95.83 | 95.83 | 95.83 | 95.83 | 94.79 | 89.58 |
| Healthy bone | 93.75 | 93.75 | 93.75 | 93.75 | 94.79 | 89.58 |
| Average | 94.79 | 94.79 | 94.79 | 94.79 | 94.79 | 89.58 |

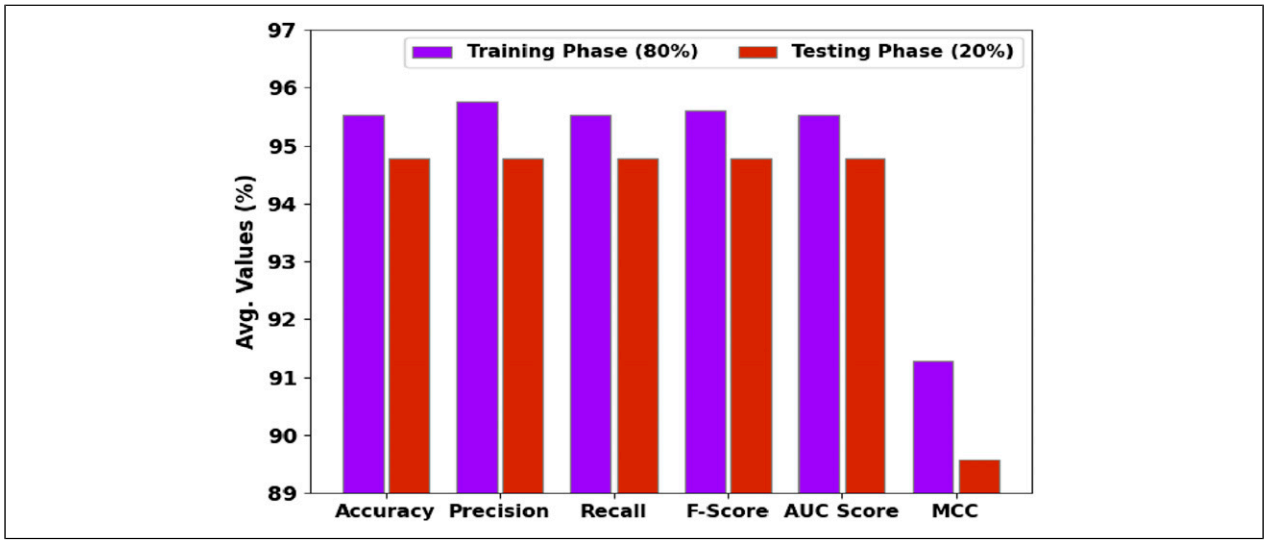


**Figure 5.** Average results of GSODL-CADBCC model on 80:20 of TRS/TSS.

**Table 4.** Bone Cancer Classification Results of GSODL-CADBCC Method on 70:30 of TRS/TSS.

| Class Labels | *Accubal* | *Precn* | *Recal* | *Fscore* | *AUCscore* | MCC |
|---|---|---|---|---|---|---|
| Training phase (70%) | | | | | | |
| Cancerous bone | 97.37 | 90.24 | 97.37 | 93.67 | 92.43 | 85.83 |
| Healthy bone | 87.50 | 96.55 | 87.50 | 91.80 | 92.43 | 85.83 |
| Average | 92.43 | 93.40 | 92.43 | 92.74 | 92.43 | 85.83 |
| Testing phase (30%) | | | | | | |
| Cancerous bone | 95.83 | 92.00 | 95.83 | 93.88 | 95.14 | 89.71 |
| Healthy bone | 94.44 | 97.14 | 94.44 | 95.77 | 95.14 | 89.71 |
| Average | 95.14 | 94.57 | 95.14 | 94.83 | 95.14 | 89.71 |

of TAC and VAC. It is also observed that the TAC and VAC reach maximum values at 200 epochs.

The training loss (TLS) and validation loss (VLS) study of the GSODL-CADBCC model is exhibited in Figure 8. It is noticed that the GSODL-CADBCC model gains reduced TLS and VLS values. In addition, it is assured that the TLS and VLS reach the least values at 200 epochs.

The selection and analysis of the loss function are crucial for training neural networks effectively. In the GSODL-CADBCC model, categorical cross-entropy loss is used because it is suitable for multi-class classification tasks. Categorical cross-entropy calculates the difference between the actual class distribution and predicted probability distribution. This is important for medical diagnosis tasks, where each class (healthy or cancerous) is

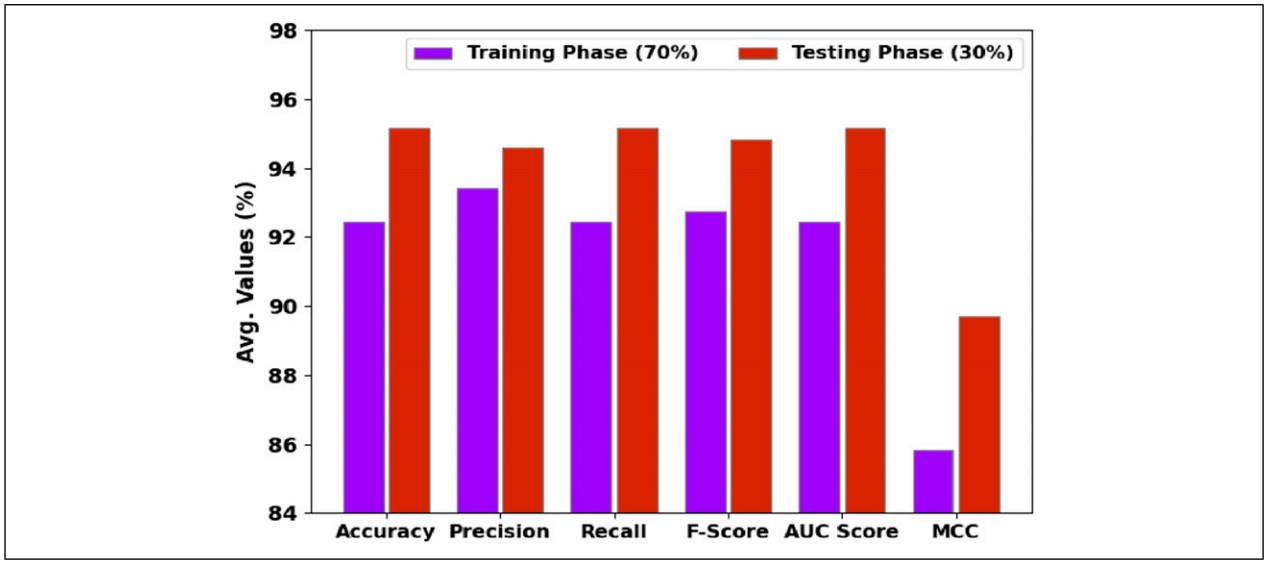


**Figure 6.** Average results of GSODL-CADBCC model on 70:30 of TRS/TSS.

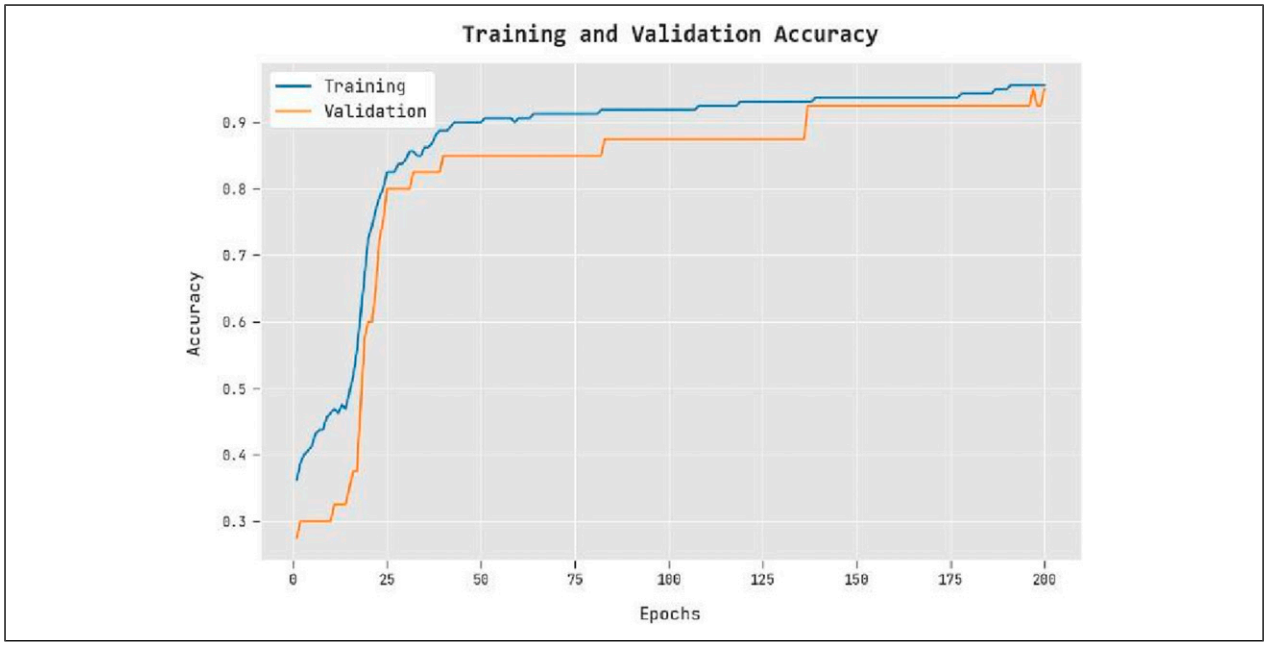


**Figure 7.** TAC and VAC study of GSODL-CADBCC model.

associated with a probability. The model outputs a probability for each class, and the categorical cross-entropy quantifies how well these probabilities match the true labels. The model's performance is validated using an 70: 30 train-test split, where 70% of the data is used for training and 30% for testing. This ensures that the model's accuracy is tested on unseen data, thus providing a measure of its generalizability.

In Figure 9, the *Precn* − *recal* study of the GSODL-CADBCC model on bone cancer classification is well studied. By looking at the results, it is guaranteed that the GSODL-CADBCC model reaches improved *precn* − *recal*values under both cancerous and healthy bone classes.

The ROC study of the GSODL-CADBCC approach is shown in Figure 10. The outcomes illustrated that the GSODL-CADBCC model is proficient with increased ROC values under both cancerous and healthy bone classes.

To exhibit the betterment of the GSODL-CADBCC method, a widespread comparison study with recent

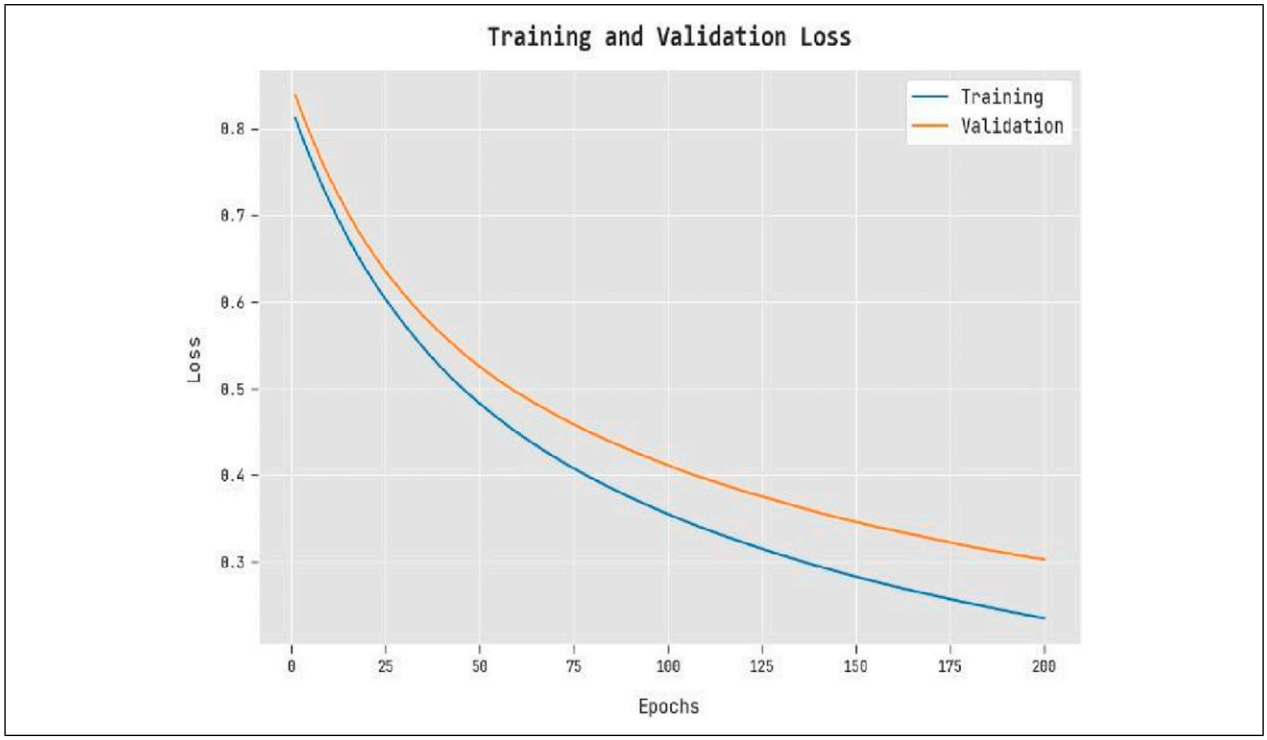


**Figure 8.** TLS and VLS study of GSODL-CADBCC model.

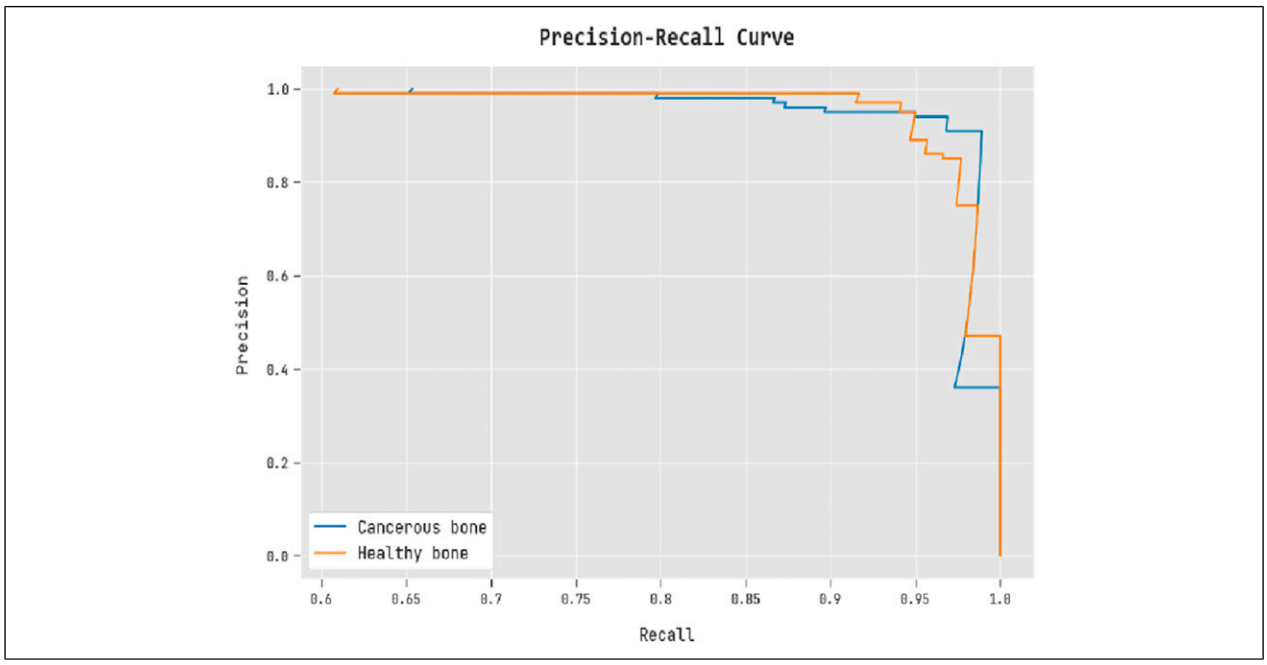


**Figure 9.** $Prec_n - reca_l$ study of GSODL-CADBCC model.

models[23,24] is given in Table 5. The result indicates that the RF model has shown poor outcomes compared to other models. Medical images, particularly X-rays for bone cancer detection, often contain complex and subtle patterns that are challenging for traditional ensemble methods like RF to capture effectively. On the other side, the inherent ability of DL models to learn complex, hierarchical representations directly from raw image data

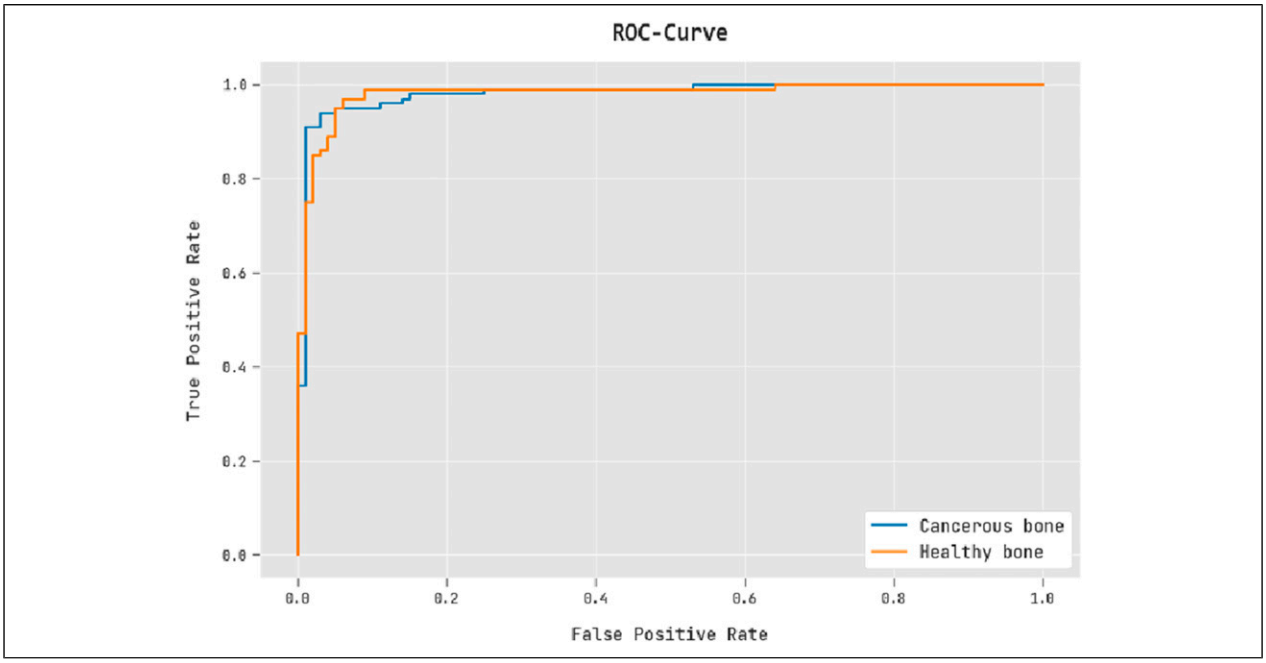


**Figure 10.** ROC study of GSODL-CADBCC model.

**Table 5.** Comparative Bone Cancer Classification Results.

| Measure | $Accu_y$ | $Prec_n$ | $Reca_l$ | *F1 score* |
|---|---|---|---|---|
| GSODL-CADBCC | 95.52 | 95.76 | 95.52 | 95.60 |
| RF algorithm | 70.39 | 76.95 | 80.06 | 78.55 |
| SVM model | 93.59 | 91.45 | 96.13 | 94.08 |
| LIO algorithm | 85.37 | 87.84 | 81.43 | 89.06 |
| FE-ML model | 92.81 | 94.27 | 90.31 | 95.03 |
| MRI-DL algorithm | 95.04 | 93.72 | 94.16 | 94.85 |

**Table 6.** Comparative CT Analysis on Bone Cancer Classification.

| Methods | Computational Time (sec) |
|---|---|
| GSODL-CADBCC | 54 |
| RF algorithm | 112 |
| SVM model | 156 |
| LIO algorithm | 204 |
| FE-ML model | 231 |
| MRI-DL algorithm | 226 |

gives them a significant advantage over traditional models like RF in medical imaging tasks. The SVM model has shown good outcomes compared to the RF model due to its effective handling of high-dimensional data and ability to find optimal decision boundaries that maximize the margin between classes. Its use of kernel functions allows it to capture complex relationships in the data, enhancing its versatility and power. In the context of the GSODL-CADBCC model, the combination of SqueezeNet feature extraction and SVM classification leverages the strengths of both DL and classical ML.

Next, the learning-based intelligent optimization (LIO) model has shown somewhat improvised performance with *accuy* of 85.37%. Followed by, the SVM, FE-ML, and MRI-DL models have accomplished moderately closer *accuy* values of 93.59%, 92.81%, and 95.04% correspondingly. However, the GSODL-CADBCC model showed promising results with a maximum *accuy* of 95.52%.

Table 6 and Figure 11 Show a comparative computation time (CT) examination of the GSODL-CADBCC method. The experimental results indicate that the FE-ML model shows poor outcomes compared to other models with CT of 231s. Next, the MRI-DL and LIO models have shown somewhat improvised performance with *accuy* of 226s and 204s correspondingly.

Followed by, the RF and SVM models have accomplished moderately closer reasonable CT of 112s and 156s correspondingly. However, the GSODL-CADBCC model showed promising results with minimal CT of 54s. These results demonstrated that the GSODL-CADBCC model can attain effective bone cancer classification performance. The validation process for the GSODL-CADBCC model is thorough, involving train-test splits and detailed evaluation metrics that ensure the model's robustness and generalizability. The use of performance metrics like precision, recall, F1-score, AUC, and MCC provides a comprehensive understanding of the model's performance. The consistent high performance across different data splits and metrics highlights the model's

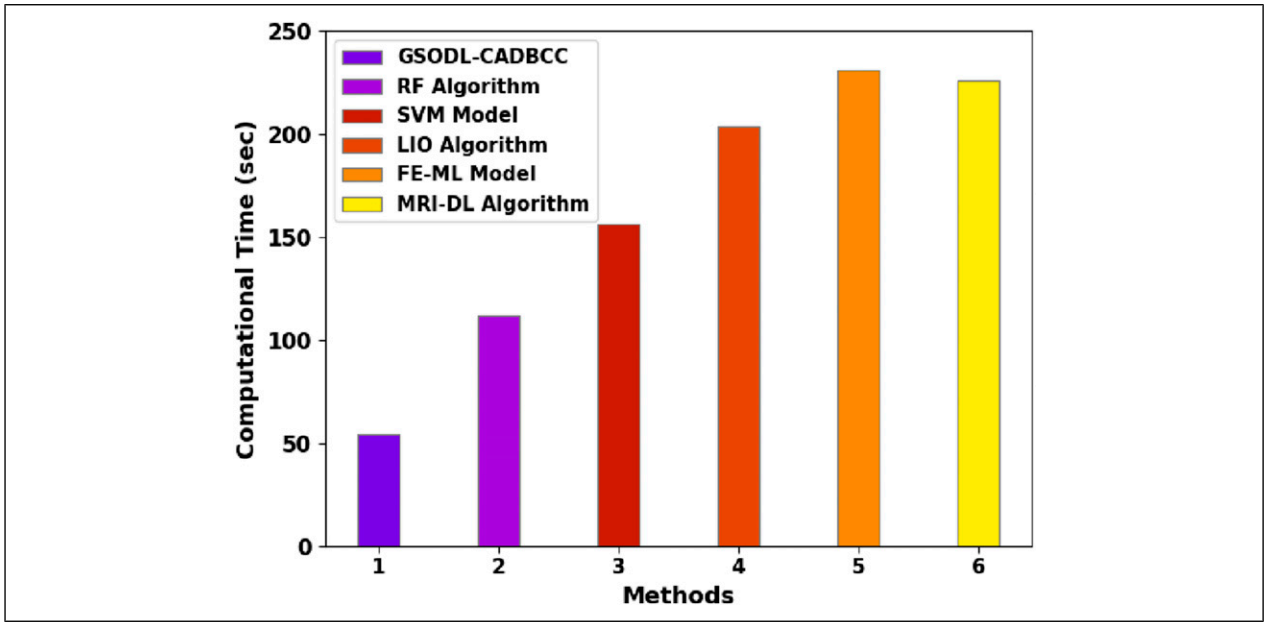


**Figure 11.** Comparative CT assessment of GSODL-CADBCC model.

effectiveness in accurately classifying bone cancer from X-ray images. This rigorous validation ensures that the GSODL-CADBCC model can be reliably used in practical medical settings, reducing the risk of diagnostic errors.

## Conclusion

In this work, an effective GSODL-CADBCC approach was introduced for automated bone cancer classification on X-ray images. The presented GSODL-CADBCC technique encompasses BF based noise removal, GSO with SqueezeNet based feature extraction, LSTM classification, and ICS based hyperparameter tuning. The design of GSO and ICS algorithms helps in the optimal selection of the hyper-parameters of the SqueezeNet and LSTM models. An extensive set of experiments was conducted to study the performance of the GSODL-CADBCC method on medical imaging datasets. Compared with recent DL models, the experimental outcomes demonstrated that the GSODL-CADBCC technique achieves promising performance on bone cancer classification. Therefore, the GSODL-CADBCC technique can be exploited for automated and accurate bone cancer classification. In the future, deep instance segmentation techniques can be included in the GSODL-CADBCC technique to improve its classification efficacy.

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### Authors' Contributions

All authors are equally contributed in the research work.

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